Seminar 4

PROBLEME TRANSFORMATA LAPLACE

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Să se determine imaginile Laplace ale următoarelor funcţii :

1. $f(t) = \sin^2 \omega t$

   R: $\mathcal{L}[f(t)](p) = \frac{2\omega^2}{p(p^2+4\omega^2)}$

2. $f(t) = (\sin t + \cos 2t)^2$

   R: $\mathcal{L}[f(t)](p) = \frac{1}{p} - \frac{p}{2(p^2+4)} + \frac{p}{2(p^2+16)} + \frac{3}{p^2+9} - \frac{1}{p^2+1}$

3. $f(t) = (t+2)^2 \mathrm{e}^{3t}$

   R: $\mathcal{L}[f(t)](p) = \frac{2}{(p-3)^3} + \frac{4}{(p-3)^2} + \frac{4}{p-3}$

4. $$f(t) = \begin{cases} \int_0^t e^{2\tau}(t-\tau)^2 d\tau, & \text{dacă } t > 0 \\ 0, & \text{dacă } t \leq 0 \end{cases}$$

   R: $\mathcal{L}[f(t)](p) = \frac{1}{p-2} \cdot \frac{2}{p^3}$

5. $f(t) = \sin at \sin bt, a, b \in \mathbb{R}$

   R: $\mathcal{L}[f(t)](p) = \frac{1}{2}\left[\frac{p}{p^2+(a-b)^2} - \frac{p}{p^2+(a+b)^2}\right]$

6. $f(t) = \int_0^t \frac{\sin \tau}{\tau} d\tau$

   R: $\mathcal{L}[f(t)](p) = \frac{\frac{\pi}{2} - \operatorname{arctg} p}{p}$

7. $f(t) = \int_0^t \frac{1-\mathrm{e}^{-\tau}}{\tau} d\tau$

   R: $\mathcal{L}[f(t)](p) = \frac{1}{p} \ln \frac{p+1}{p}$

8. $f(t) = (t-1)^2 \mathrm{e}^{t-1}$

   R: $\mathcal{L}[t^2 \mathrm{e}^t](p) = \frac{2}{(p-1)^3}$

9. $f(t) = \mathrm{sh} 2t \sin 5t$

   R: $\mathcal{L}[f(t)](p) = \frac{1}{2}\frac{5}{(p-2)^2+25} - \frac{1}{2}\frac{5}{(p+2)^2+25}$

10. $f(t) = \frac{\mathrm{e}^{bt} - \mathrm{e}^{at}}{2t\sqrt{\pi t}}$

    R: $\mathcal{L}[f(t)] = \sqrt{p-b} - \sqrt{p-a}$

Să se determine funcţiile original ale căror transformate Laplace sunt:

11. $F(p) = \frac{p^2+3p+1}{(p+1)(p+2)(p+3)}$

    R: $f(t) = -\frac{1}{2}\mathrm{e}^{-t} + \mathrm{e}^{-2t} + \frac{1}{2}\mathrm{e}^{-3t}$

12. $F(p) = \frac{2p+1}{p(p+1)}$

    R: $f(t) = 3\mathrm{e}^{t} - 1$

13. $F(p) = \frac{4p+10}{p^2-12p+32}$

    R: $f(t) = -\frac{13}{2}\mathrm{e}^{4t} + \frac{21}{2}\mathrm{e}^{8t}$

14. $F(p) = \frac{5p+1}{p^2+1}$

    R: $f(t) = 5\cos t + \sin t$

15. $F(p) = \frac{p+2}{p^2(p+3)}$

    R: $f(t) = -\frac{1}{27} + \frac{t}{9} + \frac{t^3}{3} + \frac{1}{27}\mathrm{e}^{-3t}$

16. $F(p) = \frac{1}{p^2(p-2)^2}$

    R: $f(t) = \left(\frac{t^2}{8} - \frac{t}{4} + \frac{3}{16}\right)\mathrm{e}^{2t} - \frac{t}{8} - \frac{3}{16}$

17. $F(p) = \frac{p^3+16p-24}{p^4+20p^2+64}$

    R: $f(t) = \cos 2t - \sin 2t + \frac{1}{2}\sin 4t$

18. $F(p) = \frac{2p-7}{p^2+2p+6}$

    R: $f(t) = 2\mathrm{e}^{-t}\cos\sqrt{5}t - \frac{9}{\sqrt{5}}\mathrm{e}^{-t}\sin\sqrt{5}t$

19. $F(p) = \frac{3p-14}{p^2-4p+8}$

    R: $f(t) = \mathrm{e}^{2t}(3\cos 2t - 4\sin 2t)$

20. $F(p) = \frac{\mathrm{e}^{-p}}{\sqrt{p+1}}$

    R: $f(t) = \mathrm{e}^{-(t-1)}\frac{1}{\sqrt{\pi(t-1)}}$

21. $F(p) = \frac{8\mathrm{e}^{-3p}}{p^2+4} - \frac{3p\mathrm{e}^{-2p}}{p^2-4}$

    R: $f(t) = 4\sin 2(t-3) - 3\cosh 2(t-2)$

22. $F(p) = \frac{\mathrm{e}^{-p}}{p^2-2p+5} + \frac{p\mathrm{e}^{-2p}}{p^2+9}$

    R: $f(t) = \frac{1}{2}\mathrm{e}^{t-1}\sin 2(t-1) + \cos 3(t-2)$

23. $F(p) = \frac{p\mathrm{e}^{-2p}}{p^2+3p+2}$

    R: $f(t) = 2\mathrm{e}^{-2(t-2)} - \mathrm{e}^{-(t-2)}$

24. $F(p) = \frac{27-12p}{(p+4)(p^2+9)}$

    R: $f(t) = 3\mathrm{e}^{-4t} - 3\cos 3t$

25. $F(p) = \frac{1}{2p^2-2p+5}$

R: $f(t) = \frac{1}{3}\mathrm{e}^{\frac{t}{2}} \sin\frac{3t}{2}$

26. $F(p) = \frac{3p^2-1}{(p^2+1)^2}$

R: $f(t) = (1+2t)\sin t$

27. $F(p) = \frac{5p^2-15p-11}{(p+1)(p-2)^3}$

R: $f(t) = -\frac{1}{3}\mathrm{e}^{-t} + \frac{1}{3}\mathrm{e}^{2t} + 4t\mathrm{e}^{2t} - \frac{7}{2}t^2\mathrm{e}^{2t}$

28. $F(p) = \frac{5p+1}{p^2+1}$

R: $f(t) = 5\cos t + \sin t$

29. $F(p) = \frac{\mathrm{e}^{-p}}{p^2-2p+5} + \frac{p\mathrm{e}^{-2p}}{p^2+9}$

R: $f(t) = \frac{1}{2}\mathrm{e}^{t-1}\sin 2(t-1) + \cos 3(t-2)$

30. $F(p) = \frac{1}{p(p^2+a^2)}$

R: $f(t) = \frac{1}{a^2}(1-\cos at) = \frac{2}{a^2}\sin^2\frac{at}{2}$

31. $F(p) = \frac{2p-5}{p^2-9}$

R: $f(t) = 2\mathrm{ch}3t - \frac{5}{3}\mathrm{sh}3t$

32. $F(p) = \frac{12}{4-3p} + \frac{p+1}{p^{\frac{4}{3}}}$

R: $f(t) = -4\mathrm{e}^{\frac{4t}{3}} + \frac{t^{\frac{1}{3}-1}}{\Gamma\left(\frac{1}{3}\right)} + \frac{t^{\frac{4}{3}-1}}{\Gamma\left(\frac{4}{3}\right)}$

33. $F(p) = \frac{p}{(p+1)^{\frac{5}{2}}}$

R: $f(t) = 2\mathrm{e}^{-t}\sqrt{\frac{t}{\pi}}\left(1-\frac{2}{3}t\right)$

34. $F(p) = \frac{\mathrm{e}^{-p}}{p+2} + \frac{\mathrm{e}^{-2p}}{p^2+1} + \frac{\mathrm{e}^{-3p}}{p^2+4p+5}$

R: $f(t) = \mathrm{e}^{-2(t-1)} + \sin(t-2) + \mathrm{e}^{-2(t-3)}\sin(t-3)$

35. $F(p) = \frac{15p-6}{3p^2+4p+8}$

R: $f(t) = \mathrm{e}^{-\frac{2}{3}t}(5\cos\frac{2\sqrt{5}}{3}t - \frac{8}{\sqrt{5}}\sin\frac{2\sqrt{5}}{3}t)$

36. $F(p) = \frac{3p+2}{4p^2+12p+9}$

R: $f(t) = \mathrm{e}^{-\frac{3}{2}t}\left[\frac{27}{4} - \frac{45}{8}t\right]$

37. $F(p) = \frac{2}{\sqrt[4]{16p+81}}$

R: $f(t) = \mathrm{e}^{-\frac{81}{16}t}\frac{t^{-\frac{3}{4}}}{\Gamma\left(\frac{1}{4}\right)}$

38. $F(p) = \frac{1}{(p^2+a^2)^2}, a > 0$

39. $F(p) = \frac{p}{(p^2+4)(p^2+1)}$

    R: $f(t) = \frac{1}{3}(\cos t - \cos 2t)$

40. $F(p) = \frac{p+1}{(p^2+2p+2)^2}$

    R: $f(t) = \frac{1}{2}\mathrm{e}^{-t} t \sin t$

41. $F(p) = \frac{1}{p^3+2p-3}$

    R: $f(t) = \frac{1}{5}\mathrm{e}^{t} - \frac{1}{5}\mathrm{e}^{-\frac{t}{2}} \cos \frac{\sqrt{11}}{2}t - \frac{3}{\sqrt{11}}\mathrm{e}^{-\frac{t}{2}} \sin \frac{\sqrt{11}}{2}t$

42. $F(p) = \frac{\mathrm{e}^{-p}(1-\mathrm{e}^{-p})}{p(p^2+1)}$

43. $F(p) = \frac{3p-4}{p^2-p-6}$

    R: $f(t) = \mathrm{e}^{3t} + 2\mathrm{e}^{-2t}$

Să se rezolve ecuaţiile diferenţiale :

44. $x'' + 6x' + 9x = 9\mathrm{e}^{3t}, x(0) = 0, x'(0) = 0$

    R: $x(t) = \frac{\mathrm{e}^{3t} - (1+6t)\mathrm{e}^{-3t}}{4}$

45. $y'' - 3y' + 2y = 4\mathrm{e}^{t}, y(0) = -3, y'(0) = 5$

    R: $y(t) = -7\mathrm{e}^{t} + 4t\mathrm{e}^{2t} + 4\mathrm{e}^{2t}$

46. $x'' - 2x' = \mathrm{e}^{2t} + t^2 - 1, x(0) = \frac{1}{8}, x'(0) = 1$

    R: $x(t) = -\frac{t^3}{6} - \frac{t^2}{4} + \frac{t}{4} + \frac{1}{8}\mathrm{e}^{2t}(4t+1)$

47. $x'' - 2x' + 5x = \mathrm{e}^{t} \cos 2t, x(0) = x'(0) = 1$

    R: $x(t) = \mathrm{e}^{t} \sin 2t + \frac{1}{4} t\mathrm{e}^{t} \sin 2t$

48. $y''' - 3y'' + 3y' - y = t^2\mathrm{e}^{t}, y(0) = 1, y'(0) = 0, y''(0) = -2$

    R: $y(t) = \left(1 - t - \frac{t^2}{2} + \frac{t^5}{60}\right)\mathrm{e}^{t}$

49. $x''' + x' = \sin t, x(0) = 0, x'(0) = 0, x''(0) = 0$

    R: $x(t) = 1 - \cos t - \frac{t}{2}\sin t$

50. $y'' + xy' - y = 0, y(0) = 0, y'(0) = 1$

    R: $y(x) = x$

51. $xy'' + 2y' = x - 1, y(0) = 0$

    R: $y(x) = \frac{x(x-3)}{6}$

52. $x''' - 2x'' - x' + 2x = 5\sin 2x, x(0) = 1, x'(0) = 1, x''(0) = -1$

R: $x(t) = \frac{1}{3}\mathrm{e}^{-t} + \frac{5}{12}\mathrm{e}^{2t} + \frac{1}{4}\cos 2t + \frac{1}{4}\sin 2t$

53. $x''' + 2x'' + 2x' + x = 1, x(0) = x'(0) = x''(0) = 0$

R: $x(t) = 1 - \mathrm{e}^{-t} - \frac{2}{\sqrt{3}}\mathrm{e}^{-\frac{1}{2}t}\sin\frac{\sqrt{3}}{2}t$

54. $x'' + 4x = \sin\frac{3t}{2}\sin\frac{t}{2}, x(0) = 1, x'(0) = 0$

R: $x(t) = \frac{1}{6}\cos t + \frac{5}{6}\cos 2t - \frac{1}{8}t\sin t$

55. $ty'' + y' + 4ty = 0,\ y(0) = 3, y'(0) = 0.$

56.
$$y'' + 3y' + 2y = \begin{cases} 1, & \text{dacă } 0 < t < 1 \\ 0, & \text{dacă } t > 1 \end{cases}$$

$y(0) = 0, y'(0) = 0$

57.
$$y'' + y = \begin{cases} t, & \text{dacă } 0 < t < 1 \\ 0, & \text{dacă } t > 1 \end{cases}$$

$y(0) = 0, y'(0) = 0$

Să se rezolve sistemele de ecuaţii diferenţiale:

58.
$$3x' + 2x + y' = 1$$
$$x' + 4y' + 3y = 0$$

$x(0) = y(0) = 0$

R: $x(t) = \frac{1}{2} - \frac{3}{10}\mathrm{e}^{-\frac{6}{11}t} - \frac{1}{5}\mathrm{e}^{-t}, y(t) = \frac{1}{5}(\mathrm{e}^{-t} - \mathrm{e}^{-\frac{6}{11}t})$

59.
$$x' - x + 2y = 0$$
$$x'' + 2y' = 2t - \cos 2t$$

$x(0) = 0, x'(0) = 2, y(0) = -1$

R: $x(t) = t^2 - \frac{1}{2}\sin 2t, y(t) = -t + \frac{1}{2}t^2 + \frac{1}{2}\cos 2t - \frac{1}{4}\sin 2t$

60.
$$x' = 2x - 3y$$
$$y' = y - 2x$$

$x(0) = 8, y(0) = 3$

R: $x(t) = 5\mathrm{e}^{-t} + 3\mathrm{e}^{4t}$,$y(t) = 5\mathrm{e}^{-t} - 2\mathrm{e}^{4t}$

61.
$$x' + 5x - 2y = \mathrm{e}^t$$
$$y' - x + 6y = \mathrm{e}^{2t}$$
$x(0) = 1, y(0) = -2$
R: $x(t) = \frac{7}{40}\mathrm{e}^t + \frac{1}{27}\mathrm{e}^{2t} - \frac{41}{45}\mathrm{e}^{-4t} + \frac{367}{216}\mathrm{e}^{-7t}$,
$y(t) = \frac{1}{40}\mathrm{e}^t + \frac{7}{54}\mathrm{e}^{2t} - \frac{7}{18}\mathrm{e}^{-4t} - \frac{367}{216}\mathrm{e}^{-7t}$

62.
$$x'' + x' + y'' - y = \mathrm{e}^t$$
$$x' + 2x - y' + y = \mathrm{e}^{-t}$$
$x(0) = x'(0) = 0, y(0) = y'(0) = 0$
R: $x(t) = \frac{1}{8}\mathrm{e}^t + \frac{1}{8}(2t-1)\mathrm{e}^{-t}, y(t) = \frac{3}{4}(t-1)\mathrm{e}^t - \frac{3}{4}(3t-1)\mathrm{e}^{-t}$

Să se rezolve ecuaţiile integrale:

63. $x(t) - 2\int_0^t x(\tau)d\tau = \frac{1}{9}(1 - \cos 3t)$
R: $x(t) = \frac{1}{13}\mathrm{e}^{2t} + \frac{1}{13}\cos 3t - \frac{2}{13}\sin 3t$

64. $x(t) = t + 4\int_0^t (t-\tau)x(\tau)d\tau$
R: $x(t) = \frac{1}{2}\mathrm{sh}2t$

65. $x(t) = t\cos 3t + \int_0^t \sin 3(t-\tau)x(\tau)d\tau$
R: $x(t) = 2\sin 3t - \frac{5}{\sqrt{6}}\sin\sqrt{6}t$

66. $x(t) = \cos t + \int_0^t (t-\tau)\mathrm{e}^{t-\tau}x(\tau)d\tau$
R:$x(t) = \frac{1}{5}\mathrm{e}^{2t} + \frac{4}{5}\cos t - \frac{2}{5}\sin t$

67. $x(t) - \int_0^t \mathrm{ch}2(t-\tau)\cdot x(\tau)d\tau = 4 - 4t - 8t^2$
R:$x(t) = 4 - 8t^2$

68. $x(t) = a\sin bt + c\int_0^t \sin b(t-u)\cdot x(u)du, 0 < c < b$ (ecuaţie de tip Volterra)
R:$x(t) = \frac{ab}{\sqrt{b^2-bc}}\sin\sqrt{b^2-bc}t$

69. $\int_0^t x(\tau)\frac{1}{\sqrt{t-\tau}}d\tau = 1 + t + t^2$ (Abel)
R:$x(t) = \frac{1}{\sqrt{\pi}}(\frac{t^{-\frac{1}{2}}}{\Gamma(\frac{1}{2})} + \frac{t^{\frac{1}{2}}}{\Gamma(\frac{3}{2})} + 2\frac{t^{\frac{3}{2}}}{\Gamma(\frac{5}{2})}) = \frac{1}{\pi}(\frac{1}{\sqrt{t}} + 2\sqrt{t} + \frac{8}{3}t\sqrt{t})$

70. $x(t) + 2\int_0^t x(u)du = 3\mathrm{e}^t + 2t$
R:$x(t) = 1 + \mathrm{e}^t + \mathrm{e}^{-2t}$

71. Să se rezolve ecuaţia integrodiferenţială

$$y'(t) = \int_0^t y(\tau)\cos(t-\tau)d\tau$$

cu condiţia $y(0) = 1$

R:$y(t) = 1 + \frac{t^2}{2}$

72. Să se rezolve ecuaţia diferenţială cu argumente întârziate:

$y'(t) + y(t-1) = t^2$, dacă $y(t) = 0$, $\forall t < 0$

R:$y(t) = 2\sum_{n=0}^{\infty}(-1)^n \frac{(t-n)^{n+3}}{(n+3)!}$

Să se calculeze următoarele integrale cu ajutorul transformatei Laplace:

73. $I(t) = \int_0^{\infty} \frac{\cos tx}{1+x^2}dx, t > 0$

R: $I(t) = \frac{\pi}{2}\mathrm{e}^{-t}$

74. $I(t) = \int_0^{\infty} \frac{\sin x^2}{x}dx$

R: $I = \frac{\pi}{4}$

75. $I(t) = \int_0^{\infty} \frac{\sin^2 x}{x^2}dx$

R: $I = \frac{\pi}{2}$

Rezolvări

Ecuații diferențiale

(44.) $x'' + 6x' + 9x = 9e^{3t}$, $x(0) = 0$, $x'(0) = 0$

Notăm $X(p) = \mathcal{L}[x(t)](p)$

$x''(t) \xrightarrow{\mathcal{L}} p^2X(p) - px(0) - x'(0) = p^2X(p)$

derivarea originalului

$x'(t) \xrightarrow{\mathcal{L}} pX(p) - x(0) = pX(p)$

derivarea originalului

$9e^{3t} \xrightarrow{\mathcal{L}} \frac{9}{p-3}$

$\Rightarrow$ Ecuația devine:

$$p^2X(p) + 6pX(p) + 9X(p) = \frac{9}{p-3} \iff$$

$$X(p)(p^2+6p+9) = \frac{9}{p-3} \iff X(p) = \frac{9}{p-3} \cdot \frac{1}{(p+3)^2}$$

$$\iff X(p) = -\frac{1}{4(p+3)} - \frac{3}{2(p+3)^2} + \frac{1}{4(p-3)} \iff$$

$$x(t) = \mathcal{L}\left[-\frac{1}{4(p+3)}\right](t) + \mathcal{L}\left[-\frac{3}{2(p+3)^2}\right](t) + \mathcal{L}\left[\frac{1}{4(p-3)}\right](t)$$

$$\iff x(t) = -\frac{1}{4}e^{-3t} - \frac{3}{2}e^{-3t} \cdot t + \frac{1}{4}e^{3t} \iff$$

$$\boxed{x(t) = \frac{e^{3t} - e^{-3t}(6t+1)}{4}}$$

(45.) $y'' - 3y' + 2y = 4e^t$, $y(0) = -3$, $y'(0) = 5$

Notăm $Y(p) = \mathcal{L}[y(t)](p)$

$y''(t) \xrightarrow{\mathcal{L}} p^2Y(p) - py(0) - y'(0) = p^2Y(p) + 3p - 5$

derivarea originalului

$y'(t) \xrightarrow{\mathcal{L}} pY(p) - y(0) = pY(p) + 3$

derivarea originalului

$4e^t \xrightarrow{\mathcal{L}} 4 \cdot \frac{1}{p-1}$

$\Rightarrow$ Ecuația devine:

$$p^2Y(p) + 3p - 5 - 3(pY(p)+3) + 2Y(p) = 4 \cdot \frac{1}{p-1} \iff$$

$$Y(p)(p^2 - 3p + 2) + 3p - 5 - 9 = 4 \cdot \frac{1}{p-1} \iff Y(p)(p-1)(p-2) =$$

$$4 \cdot \frac{1}{p-1} + 14 - 3p \iff Y(p) = \frac{1}{(p-1)(p-2)} \left(4 \cdot \frac{1}{p-1} + 14 - 3p \right)$$

$$\iff Y(p) = \frac{4}{(p-1)^2(p-2)} + \frac{14-3p}{(p-1)(p-2)} \iff$$

$$Y(p) = -\frac{4}{p-1} - \frac{4}{(p-1)^2} + \frac{4}{p-2} - \frac{11}{p-1} + \frac{8}{p-2} \iff$$

$$Y(p) = -\frac{15}{p-1} + \frac{12}{p-2} - \frac{4}{(p-1)^2}$$

derivare a originalului $\iff$

$$y(t) = -15e^t + 12e^{2t} - 4e^{-t} \cdot t$$

(46) $x'' - 2x' = e^{2t} + t^2 - 1$, $x(0) = \frac{1}{8}$, $x'(0) = 1$

Notăm $X(p) = \mathcal{L}[x(t)](p)$

Conform teoremei derivării originalului:

$$x'(t) \xrightarrow{\mathcal{L}} pX(p) - x(0) = pX(p) - \frac{1}{8}$$

$$x''(t) \xrightarrow{\mathcal{L}} p^2X(p) - px(0) - x'(0) = p^2X(p) - \frac{p}{8} - 1$$

$$e^{2t} + t^2 - 1 \xrightarrow{\mathcal{L}} \frac{1}{p-2} + \frac{2}{p^3} - \frac{1}{p}$$

$\Rightarrow$ Ecuația devine:

$$p^2X(p) - \frac{p}{8} - 1 - 2\left(pX(p) - \frac{1}{8}\right) = \frac{1}{p-2} + \frac{2}{p^3} - \frac{1}{p}$$

$$\iff X(p)(p^2 - 2p) - \frac{p}{8} - 1 + \frac{1}{4} = \frac{1}{p-2} + \frac{2}{p^3} - \frac{1}{p} \iff$$

$\Leftrightarrow X(p)\,p(p-2) = \frac{1}{p-2} + \frac{2}{p^3} - \frac{1}{p} + \frac{p}{8} + \frac{3}{4} \Leftrightarrow$

$X(p) = \frac{1}{p(p-2)^2} + \frac{2}{p^4(p-2)} - \frac{1}{p^2(p-2)} + \frac{1}{8(p-2)} + \frac{3}{p(p-2)}$

$\Leftrightarrow X(p) = \frac{1}{4p} - \frac{1}{4(p-2)} + \frac{1}{2(p-2)^2} - \frac{1}{8p} - \frac{1}{4p^2} - \frac{1}{2p^3} -$

$\frac{1}{p^4} + \frac{1}{8(p-2)} - \frac{1}{4p} + \frac{1}{2p^2} + \frac{1}{4(p-2)} + \frac{1}{8(p-2)} - \frac{1}{8(p-2)} + \frac{1}{8p}$

$\Leftrightarrow X(p) = \frac{-1}{p^4} + \frac{1}{p^3}\left(-\frac{1}{2}\right) + \frac{1}{p^2}\left(-\frac{1}{4}+\frac{1}{2}\right) + \frac{1}{p}\left(\cancel{\frac{1}{4}} - \cancel{\frac{1}{8}} + \cancel{\frac{1}{4}} + \cancel{\frac{1}{8}}\right) + \frac{1}{p-2}\left(-\cancel{\frac{1}{4}} + \frac{1}{8} + \cancel{\frac{1}{4}} + \cancel{\frac{1}{8}} - \cancel{\frac{1}{8}}\right) + \frac{1}{(p-2)^2}\cdot\frac{1}{2} \Leftarrow$

$x(t) = -\frac{t^3}{6} - \frac{1}{2}\cdot\frac{t^2}{2} + t\cdot\frac{1}{4} + e^{2t}\cdot\frac{1}{8} + \frac{1}{2}\,e^{2t}\cdot t \Leftrightarrow$

$x(t) = -\frac{t^3}{6} - \frac{1}{4}t^2 + t\,\frac{1}{4} + e^{2t}\,\frac{1}{8} + \frac{1}{2}\cdot e^{2t}\,t$

47. $x'' - 2x' + 5x = e^t \cos 2t$, $x(0) = x'(0) = 1$

Notăm $X(p) = \mathcal{L}[x(t)](p)$

Conform teoremei derivării originalului avem:

$x'(t) \xrightarrow{\mathcal{L}} pX(p) - x(0) = pX(p) - 1$

$x''(t) \xrightarrow{\mathcal{L}} p^2X(p) - px(0) - x'(0) = p^2X(p) - p - 1$

$e^t\cos 2t \xrightarrow[\text{deplasare}]{\mathcal{L}} \mathcal{L}[\cos 2(t-1)] \longrightarrow \frac{p-1}{(p-1)^2+4}$

Ecuația devine:

$(p^2X(p) - p - 1) - 2(pX(p) - 1) + 5X(p) = \frac{p-1}{(p-1)^2+4}$

$\Leftrightarrow X(p)(p^2 - 2p + 5) - p - 1 + 2 = \frac{p-1}{(p-1)^2+4}$

$\Leftrightarrow X(p)(p^2-4p+5) = \frac{p-1}{(p-1)^2+4} + p-1 \Leftrightarrow$

$X(p)((p-1)^2+4) = \frac{p-1}{(p-1)^2+4} + p-1 \Leftrightarrow$

$X(p) = \frac{p-1}{((p-1)^2+4)^2} + \frac{p-1}{(p-1)^2+4} \overset{\text{deplasare}}{\underset{\text{derivarea originalu}}{\Leftrightarrow}} x(t) = e^t\cos 2t +$

$e^t \cos 2t \cdot t \cdot \frac{1}{4}$

(63) **Ecuații integrale**

$x(t) - 2\int_0^t x(\tau)\,d\tau = \frac{1}{9}(1-\cos 3t)$

Notăm $x(t) \xrightarrow{\mathcal{L}} X(p)$

$2\int_0^t x(\tau)\,d\tau \xrightarrow[\text{integrarea originalului}]{\mathcal{L}} 2\cdot\frac{X(p)}{p}$

$\frac{1}{9}(1-\cos 3t) \xrightarrow[\text{liniaritate}]{\mathcal{L}} \frac{1}{9}\mathcal{L}[u(t)](p) - \frac{1}{9}\mathcal{L}[\cos 3t](p)$

$= \frac{1}{9}\cdot\frac{1}{p} - \frac{1}{9}\cdot\frac{p}{p^2+9} = \frac{1}{9}\left(\frac{1}{p} - \frac{p}{p^2+9}\right) \Rightarrow$

Ecuația devine:

$X(p) - 2\frac{X(p)}{p} = \frac{1}{9}\left(\frac{1}{p} - \frac{p}{p^2+9}\right) \overset{\cdot p}{\Rightarrow}$

$pX(p) - 2X(p) = \frac{1}{9}\left(1 - \frac{p^2}{p^2+9}\right) \Rightarrow$

$X(p)(p-2) = \frac{1}{9}\cdot\frac{p^2+9-p^2}{p^2+9} \Rightarrow X(p)(p-2) = \frac{1}{p^2+9}$

$$\Rightarrow \quad X(p) = \frac{1}{(p-2)(p^2+9)} \quad \Rightarrow \quad X(p) = \frac{1}{13(p-2)} - \frac{p+2}{13(p^2+9)} =$$

$$= \frac{1}{13(p-2)} - \frac{p}{13(p^2+9)} - \frac{2}{13(p^2+9)} \Rightarrow$$

$$x(t) = \frac{1}{13} e^{2t} - \frac{1}{13} \cos 3t - \frac{2}{39} \sin 3t$$